

Superconformal blocks on a plumbing graph

Supplementary implementation notes

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A superconformal block is a sum of three-point forms sewn with inverse Gram matrices. We compute it by adjoining a free fermion, reducing the enlarged block to a sum of products of two Virasoro blocks, and removing the fermion factor by convolution. The challenge is to keep the operator order and local normalizations consistent through those three steps.

We first specify the BPZ pairings, ground values, and contour branches. Their tensor product gives the enlarged PBW sewing and its convolution. The double-Virasoro formula evaluates its enlarged side. Finally, the Ramond cycle identities identify the sector on which recovery is possible; one external $\Theta\psi$ insertion supplies a missing sector. For Mercedes with NS spokes and a Ramond rim, this requires at most one inserted sphere, while the central all-NS sphere stays sewn in place.

We keep the notation of the SCblock notes and the draft. In particular, K denotes the operator-reordering exponent and $\star$ its associated convolution. The formulas below display the state arguments and local signs explicitly; no separate state aliases or parity-transformation operators are needed.

1 States, three-point forms, and physical sewing

1.1 The channel and its labels

A plumbing channel $\hat{\Gamma}$ specifies the NS/R edges, local coordinates, and spin lifts $\eta_e = \pm 1$. Each trivalent sphere is either all-NS or NS–R–R. Fix the slots $(\infty, 1, 0)$, numbered 1, 2, 3; at an NS–R–R sphere slot 1 is NS. Also fix an order of the spheres, the internal edges, the two ends of each edge, and the external punctures. These choices specify the operator order as well as the sewing. Changing a slot order requires transporting its states, coordinates, and plumbing maps together. A self-loop has two distinct ends at the same vertex.

Subscripts $(e, 1), (e, 2)$ label the ends of an internal edge, while $(v, 1), (v, 2), (v, 3)$ label the slots of a sphere. They refer to the same punctures: for example, $A_{v,i}$ is the PBW word at the end of the edge incident to that slot. Throughout, edge products and internal-state sums run only over internal edges unless stated otherwise. External states are fixed. Products restricted by $e : \text{NS}$, $e : \text{R}$, or $v : \text{NS}, \text{R}, \text{R}$ specify the corresponding edge or vertex types, without introducing separate sets for each type.

For a connected trivalent channel the genus is the number of internal edges minus the number of vertices plus one. The elementary unstable sphere pairings are treated separately; an unpunctured torus can first be given an identity puncture. Equivalent local spin frames do not count as additional spin structures.

1.2 The PBW basis and local forms

An NS PBW state is $\mathbf{L}_{-A}\phi$, and a Ramond state is $\mathbb{L}_{-A}w^\alpha$, where the word contains ordered L_{-n} and G_{-r} with positive mode indices. The G indices are half-integer in NS and integer in R. As in the

draft, $|A|$ is the sum of the mode indices, whereas A in a sign counts the G operators modulo two. We take NS highest-weight vectors even; odd NS states are their descendants. The physical Ramond grounds are w^+ even and w^- odd. Thus

$$\epsilon = \begin{cases} A, & \text{NS,} \\ A + |\alpha|, & \text{R,} \end{cases} \quad | + | = 0, \quad | - | = 1. \quad (1.1)$$

Every parity is represented by zero or one. Parity sums are modulo two unless a different meaning is explicitly specified. The Ramond parameter obeys $\beta^2 = c/24 - h$, where c is the physical superconformal central charge, and $G_0 w^\pm = i\beta e^{\mp i\pi/4} w^\mp$. Use bilinear BPZ conjugation

$$L_n^\dagger = L_{-n}, \quad G_r^\dagger = G_{-r}, \quad \psi_r^\dagger = -\psi_{-r}. \quad (1.2)$$

Normalize the NS primary to norm one. With the stated G_0 action, its adjoint fixes the physical Ramond ground metric to $\text{diag}(1, i)$. Coefficients are not complex-conjugated.

Lower Gram indices denote pairings and upper indices their inverses:

$$\begin{aligned} (\mathbf{G}_h)_{AB} &= \langle \mathbf{L}_{-A} \phi_h | \mathbf{L}_{-B} \phi_h \rangle, \\ (\mathbb{G}_\beta)_{A\alpha, B\gamma} &= \langle \mathbb{L}_{-A} w_\beta^\alpha | \mathbb{L}_{-B} w_\beta^\gamma \rangle. \end{aligned} \quad (1.3)$$

They pair equal levels and equal parities, so a nonzero edge contraction has one common level and one common ϵ_e at its two ends. These are bilinear BPZ pairings, without complex conjugating coefficients. We assume generic weights for which the inverse Gram matrices exist. Degenerate modules require their null-state quotient or a specified regular limit.

At a vertex the label is $a_v = 0, 1$ for all-NS, or $f_v = 0, 1$ and $\eta_v = \pm 1$ for NS-R-R. The vertex sign η_v labels a three-point form; the edge sign η_e labels sewing. The forms at $(\infty, 1, 0)$ are normalized by

$$\begin{aligned} \rho_0(\phi_1, \phi_2, \phi_3) &= \rho_1(\phi_1, G_{-1/2} \phi_2, \phi_3) = 1, \\ \rho_0^{(\eta)}(\phi_1, w_2^+, w_3^+) &= \rho_1^{(\eta)}(\phi_1, w_2^+, w_3^-) = 1, \\ \rho_0^{(\eta)}(\phi_1, w_2^-, w_3^-) &= \eta, \quad \rho_1^{(\eta)}(\phi_1, w_2^-, w_3^+) = -i\eta. \end{aligned} \quad (1.4)$$

Unlisted Ramond ground components vanish. Superconformal Ward identities in this slot order determine their descendants. Nonzero components obey

$$\sum_i \epsilon_{v,i} = \begin{cases} a_v, & (\text{NS}, \text{NS}, \text{NS}), \\ f_v, & (\text{NS}, \text{R}, \text{R}). \end{cases} \quad (1.5)$$

Equivalently, the descendant and ground labels obey $A+B+C = a$ in all-NS and $A+B+C+|\alpha|+|\gamma| = f$ in NS-R-R. These constraints are used below to simplify the signs.

1.3 The operator-reordering sign and the block

PBW sewing initially gives consecutive pairs of operators, one pair per internal edge, followed by the external operators. To evaluate the local three-point forms, reorder them into consecutive vertex triples. The exponent K records the odd exchanges:

$$K(\epsilon) = \sum_{\substack{(v,i) \text{ precedes } (w,j) \text{ in edge-pair order} \\ (w,j) \text{ precedes } (v,i) \text{ in vertex order}}} \epsilon_{v,i} \epsilon_{w,j}. \quad (1.6)$$

The external punctures are included in the first order after all internal pairs. Each reversed pair is counted once. For example, in theta the orders $(1_L, 1_R, 2_L, 2_R, 3_L, 3_R)$ and $(1_L, 2_L, 3_L, 1_R, 2_R, 3_R)$ give $K = \epsilon_1\epsilon_2 + \epsilon_1\epsilon_3 + \epsilon_2\epsilon_3$. The same prescription includes self-loops and external odd states without referring to crossings in a drawing.

The physical block now follows by inserting each inverse Gram matrix between its two vertex forms:

$$\begin{aligned} \mathbf{F}_{\hat{\Gamma}} = & \sum_{\{A, \alpha\}} (-1)^{K(\epsilon)} \prod_{e: \text{NS}} q_e^{|A_{e,1}|} \eta_e^{\epsilon_e} \mathbf{G}_{h_e}^{A_{e,1} A_{e,2}} \prod_{e: \text{R}} q_e^{|A_{e,1}|} \eta_e^{\epsilon_e} \mathbb{G}_{\beta_e}^{A_{e,1} \alpha_{e,1}, A_{e,2} \alpha_{e,2}} \\ & \times \prod_{v: \text{NS}, \text{NS}, \text{NS}} \rho_{a_v}(\mathbf{L}_{-A_{v,1}} \phi_{v,1}, \mathbf{L}_{-A_{v,2}} \phi_{v,2}, \mathbf{L}_{-A_{v,3}} \phi_{v,3}) \\ & \times \prod_{v: \text{NS}, \text{R}, \text{R}} \rho_{f_v}^{(\eta_v)}(\mathbf{L}_{-A_{v,1}} \phi_{v,1}, \mathbb{L}_{-A_{v,2}} w_{v,2}^{\alpha_{v,2}}, \mathbb{L}_{-A_{v,3}} w_{v,3}^{\alpha_{v,3}}). \end{aligned} \quad (1.7)$$

The sum includes both physical endpoint words on every internal edge and both Ramond ground labels where applicable. Equal-level and equal-parity constraints are enforced by the inverse Gram entries. The local constraints are enforced by the vertex forms. The powers $q_e^{h_e}$ are excluded from this descendant-normalized block and restored in a correlator. External finite linear combinations are treated by linearity.

2 The local signs and the enlarged block

2.1 The auxiliary basis and its BPZ pairing

Adjoin an auxiliary fermion, with $\{\psi_r, \psi_s\} = \delta_{r+s,0}$ and $\psi_r^\dagger = -\psi_{-r}$. The minus sign in this BPZ conjugation is required for the two Virasoro generators to obey $(L_n^{(j)})^\dagger = L_{-n}^{(j)}$. Write $\Psi_{-A} = \psi_{-r_1} \cdots \psi_{-r_m}$ with $r_1 > \cdots > r_m > 0$. The modes are half-integer in NS and integer in R; A in a sign counts these modes, and $|A|$ is their sum. Zero modes are represented by the ground labels, not included in Ψ_{-A} . The auxiliary states and parities are

	state	ϵ'	
NS	$\Psi_{-A} \mathbf{1}$	$A \bmod 2$	$\mathbf{a} \in \{0, 1\}$.
R	$\Psi_{-A} u^{\mathbf{a}}$	$(A + \mathbf{a}) \bmod 2$	

(2.1)

Here ϵ continues to denote physical parity, and ϵ' denotes auxiliary parity; the tensor state has parity $\epsilon + \epsilon'$. Choose $\psi_0 u^{\mathbf{a}} = u^{1-\mathbf{a}}/\sqrt{2}$ and ground norms $1, -1$. The oscillator anticommutators then give

$$\langle\langle \Psi_{-A} u^{\mathbf{a}} | \Psi_{-B} u^{\mathbf{b}} \rangle\rangle = \begin{cases} (-1)^{A+\mathbf{a}}, & (A, \mathbf{a}) = (B, \mathbf{b}), \\ 0, & \text{otherwise.} \end{cases} \quad (2.2)$$

For NS replace the grounds by $\mathbf{1}$ and set $\mathbf{a} = 0$. Thus the inverse auxiliary Gram entry is also $(-1)^{\epsilon'}$. On the tensor basis we use the product of the physical and auxiliary BPZ pairings, without an additional crossing sign. This local convention is separate from the global operator permutation counted by K .

The normalized auxiliary three-point form is fixed by

$$\rho_{\mathbf{F}}(\mathbf{1}, \mathbf{1}, \mathbf{1}) = \rho_{\mathbf{F}}(\mathbf{1}, u^0, u^0) = 1, \quad \rho_{\mathbf{F}}(\mathbf{1}, u^1, u^1) = i, \quad (2.3)$$

and the fermion Ward identities. It vanishes unless the three auxiliary parities sum to zero. The BPZ signs of bra descendants are already included in $\rho_{\mathbf{F}}$; they must not be added a second time.

2.2 Contour branches and Ward identities

In these identities x_i is a state at the indicated puncture, physical for $\rho_f^{(\eta)}$ and auxiliary for ρ_F . The symbols ϵ_2, ϵ'_2 denote the corresponding middle-state parities. Choose the square root with expansions

$$\sqrt{z(z-1)} = \begin{cases} z\sqrt{1-z^{-1}}, & z \sim \infty, \\ \sqrt{z-1}\sqrt{1+(z-1)}, & z \sim 1, \\ i\sqrt{z}\sqrt{1-\bar{z}}, & z \sim 0. \end{cases} \quad (2.4)$$

The last expression is the upper-lip continuation of the branch cut between the Ramond punctures. For integers m, n , apply the residue theorem to the current correlator multiplied by $z^m(z-1)^n\sqrt{z(z-1)}$ for G , and by $z^m(z-1)^n/\sqrt{z(z-1)}$ for ψ . At infinity use the stated BPZ adjoints. Thus the supercurrent identity is

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n + \frac{1}{2}}{j} \rho_f^{(\eta)}(G_{j-m-n-\frac{1}{2}}x_1, x_2, x_3) \\ &= \sum_{j \geq 0} \binom{m + \frac{1}{2}}{j} \rho_f^{(\eta)}(x_1, G_{n+j}x_2, x_3) \\ &+ i(-1)^{\epsilon_2+n} \sum_{j \geq 0} (-1)^j \binom{n + \frac{1}{2}}{j} \rho_f^{(\eta)}(x_1, x_2, G_{m+j}x_3), \end{aligned} \quad (2.5)$$

where ϵ_2 is the parity of the middle physical state x_2 . The auxiliary identity is

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n - \frac{1}{2}}{j} \rho_F(\psi_{j-m-n+\frac{1}{2}}x_1, x_2, x_3) \\ &= - \sum_{j \geq 0} \binom{m - \frac{1}{2}}{j} \rho_F(x_1, \psi_{n+j}x_2, x_3) \\ &+ i(-1)^{\epsilon'_2+n} \sum_{j \geq 0} (-1)^j \binom{n - \frac{1}{2}}{j} \rho_F(x_1, x_2, \psi_{m+j}x_3), \end{aligned} \quad (2.6)$$

with ϵ'_2 the middle auxiliary parity. Every sum terminates on finite-level states. No three-point parity label occurs in the contour sign: the sign counts the state that the current passes at $z = 1$.

For example, on Ramond ground states the first identity gives

$$\begin{aligned} \rho_f^{(\eta)}(G_{-1/2}\phi_1, w_2^\alpha, w_3^\gamma) &= \rho_f^{(\eta)}(\phi_1, G_0w_2^\alpha, w_3^\gamma) \\ &+ i(-1)^{|\alpha|} \rho_f^{(\eta)}(\phi_1, w_2^\alpha, G_0w_3^\gamma). \end{aligned} \quad (2.7)$$

The auxiliary identity with $m = n = 0$ and grounds $(\mathbf{1}, u^0, u^1)$ forces $\rho_F(\mathbf{1}, u^1, u^1) = i\rho_F(\mathbf{1}, u^0, u^0)$. Thus the remaining auxiliary i has a direct zero-mode Ward explanation.

The Virasoro Ward identities use the ordinary residue theorem with $z^m(z-1)^nT(z)$, or explicitly, for either three-point form,

$$\begin{aligned}
& \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho(L_{j-m-n+1} x_1, x_2, x_3) \\
&= \sum_{j \geq 0} \binom{m}{j} \rho(x_1, L_{n+j-1} x_2, x_3) \\
&+ (-1)^n \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho(x_1, x_2, L_{m+j-1} x_3).
\end{aligned} \tag{2.8}$$

For all-NS vertices the currents are single-valued: omit the square-root multiplier and use the same BPZ and operator-ordering prescription.

2.3 The enlarged three-point form

Order each tensor state with the auxiliary factor first. Factoring its matrix element moves the middle physical operators past the ket auxiliary operators. The bra uses the product BPZ pairing already specified. Therefore

$$\begin{aligned}
& \hat{\rho}_a(\Psi_{-A} \mathbf{1} \otimes \mathbf{L}_{-A} \phi_1, \Psi_{-B} \mathbf{1} \otimes \mathbf{L}_{-B} \phi_2, \Psi_{-C} \mathbf{1} \otimes \mathbf{L}_{-C} \phi_3) \\
&= (-1)^{BC} \rho_F(\Psi_{-A} \mathbf{1}, \Psi_{-B} \mathbf{1}, \Psi_{-C} \mathbf{1}) \rho_a(\mathbf{L}_{-A} \phi_1, \mathbf{L}_{-B} \phi_2, \mathbf{L}_{-C} \phi_3),
\end{aligned} \tag{2.9}$$

$$\begin{aligned}
& \hat{\rho}_f^{(\eta)}(\Psi_{-A} \mathbf{1} \otimes \mathbf{L}_{-A} \phi_1, \Psi_{-B} u^b \otimes \mathbb{L}_{-B} w_2^\alpha, \Psi_{-C} u^c \otimes \mathbb{L}_{-C} w_3^\gamma) \\
&= (-1)^{(B+|\alpha|)(C+c)} \rho_F(\Psi_{-A} \mathbf{1}, \Psi_{-B} u^b, \Psi_{-C} u^c) \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma).
\end{aligned} \tag{2.10}$$

Both signs are $(-1)^{\epsilon_2 \epsilon'_3}$. There is no further local phase. The ground values, Ward identities, and tensor pairing specified above are parts of this single definition.

2.4 Definition of the enlarged block

We define the enlarged block by sewing the three-point forms (2.9) and (2.10) in the tensor-product PBW basis. On each internal edge e , sum independently over the two physical words $A_{e,1}, A_{e,2}$ and, for a Ramond edge, the two physical ground labels $\alpha_{e,1}, \alpha_{e,2} \in \{+, -\}$. The auxiliary Gram matrix is diagonal, so there is a single auxiliary word A_e shared by the two ends, together with a single ground label $a_e \in \{0, 1\}$ on each Ramond edge. With the external states and the vertex labels fixed, the definition is

$$\begin{aligned}
\hat{\mathbf{F}}_{\hat{\Gamma}} &= \sum_{\{A, \alpha, A, a\}} (-1)^{K(\epsilon + \epsilon')} \prod_e q_e^{|A_{e,1}| + |A_e|} \eta_e^{\epsilon_e + \epsilon'_e} (-1)^{\epsilon'_e} \\
&\times \prod_{e: \text{NS}} \mathbf{G}_{h_e}^{A_{e,1} A_{e,2}} \prod_{e: \text{R}} \mathbb{G}_{\beta_e}^{A_{e,1} \alpha_{e,1}, A_{e,2} \alpha_{e,2}} \\
&\times \prod_{v: \text{NS}, \text{NS}, \text{NS}} \hat{\rho}_{a_v}(\Psi_{-A_{v,1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,1}} \phi_{v,1}, \\
&\quad \Psi_{-A_{v,2}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,2}} \phi_{v,2}, \Psi_{-A_{v,3}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,3}} \phi_{v,3}) \\
&\times \prod_{v: \text{NS}, \text{R}, \text{R}} \hat{\rho}_{f_v}^{(\eta_v)}(\Psi_{-A_{v,1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,1}} \phi_{v,1}, \\
&\quad \Psi_{-A_{v,2}} u^{\mathbf{a}_{v,2}} \otimes \mathbb{L}_{-A_{v,2}} w_{v,2}^{\alpha_{v,2}}, \Psi_{-A_{v,3}} u^{\mathbf{a}_{v,3}} \otimes \mathbb{L}_{-A_{v,3}} w_{v,3}^{\alpha_{v,3}}).
\end{aligned} \tag{2.11}$$

Here ϵ_e and ϵ'_e are the physical and auxiliary parities defined in (1.1) and (2.1). The labels with subscripts (v, i) are those at slot i of sphere v ; the auxiliary labels are inherited from its incident

edge. Every edge product and every summed label is internal. At an external puncture, insert its prescribed tensor-PBW state into the corresponding argument of $\hat{\rho}$; finite linear combinations are treated by linearity. The inverse physical Gram matrices impose equal physical levels and parities at the two ends of each edge. The local forms impose the vertex parity constraints. As for the physical block, the powers from the intermediate ground-state weights are omitted.

Each factor in (2.11) has a direct sewing meaning. The power of q_e is the sum of the physical and auxiliary descendant levels. The spin lift η_e acts on their total parity, while $(-1)^{\epsilon'_e}$ is the inverse auxiliary Gram entry (2.2). Finally, $K(\epsilon + \epsilon')$ counts the exchanges of the total tensor-state parities in the specified operator order. There is no additional NS-R-R vertex or NS-edge weight.

2.5 The fermion block

Sew the auxiliary forms alone in the same operator order:

$$\begin{aligned} \mathbf{F}_{\mathbf{F},\hat{\Gamma}} &= \sum_{\{\mathbf{A}_e, \mathbf{a}_e\}} (-1)^{K(\epsilon')} \prod_e q_e^{|\mathbf{A}_e|} (-\eta_e)^{\epsilon'_e} \\ &\quad \times \prod_{v:\text{NS,NS,NS}} \rho_{\mathbf{F}}(\Psi_{-\mathbf{A}_{v,1}} \mathbf{1}, \Psi_{-\mathbf{A}_{v,2}} \mathbf{1}, \Psi_{-\mathbf{A}_{v,3}} \mathbf{1}) \\ &\quad \times \prod_{v:\text{NS,R,R}} \rho_{\mathbf{F}}(\Psi_{-\mathbf{A}_{v,1}} \mathbf{1}, \Psi_{-\mathbf{A}_{v,2}} u^{\mathbf{a}_{v,2}}, \Psi_{-\mathbf{A}_{v,3}} u^{\mathbf{a}_{v,3}}). \end{aligned} \quad (2.12)$$

One auxiliary word and, on a Ramond edge, one ground label are shared by the two endpoints. External states are fixed; use $\mathbf{1}$ at each original external NS puncture and u^0 at each original external R puncture for recovery. The factor $(-1)^{\epsilon'_e}$ is the inverse metric. Terms violating auxiliary parity conservation vanish.

For theta, $\epsilon'_1 + \epsilon'_2 + \epsilon'_3 = 0$, so the three inverse-metric signs multiply to one. Hence

$$\begin{aligned} \mathbf{F}_{\mathbf{F},\theta} &= \sum_{\substack{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{b}, \mathbf{c} \\ \mathbf{A} + \mathbf{B} + \mathbf{C} + \mathbf{b} + \mathbf{c} = 0 \pmod{2}}} (-1)^{\epsilon'_1 \epsilon'_2 + \epsilon'_1 \epsilon'_3 + \epsilon'_2 \epsilon'_3} q_1^{|\mathbf{A}|} q_2^{|\mathbf{B}|} q_3^{|\mathbf{C}|} \eta_1^{\epsilon'_1} \eta_2^{\epsilon'_2} \eta_3^{\epsilon'_3} \\ &\quad \times [\rho_{\mathbf{F}}(\Psi_{-\mathbf{A}} \mathbf{1}, \Psi_{-\mathbf{B}} u^{\mathbf{b}}, \Psi_{-\mathbf{C}} u^{\mathbf{c}})]^2, \end{aligned} \quad (2.13)$$

where $(\epsilon'_1, \epsilon'_2, \epsilon'_3) = (\mathbf{A}, \mathbf{B} + \mathbf{b}, \mathbf{C} + \mathbf{c})$ modulo two.

3 Convolution

A term in the physical block has powers r_e and parities ϵ_e ; a term in the auxiliary block has powers s_e and parities ϵ'_e . Their product adds levels on each edge and adds parities modulo two. There are only two sources of a mixed sign. At a reversed pair of punctures, subtracting the separate physical and auxiliary signs from the tensor-state sign leaves

$$(\epsilon_{v,i} + \epsilon'_{v,i})(\epsilon_{w,j} + \epsilon'_{w,j}) - \epsilon_{v,i} \epsilon_{w,j} - \epsilon'_{v,i} \epsilon'_{w,j} = \epsilon_{v,i} \epsilon'_{w,j} + \epsilon'_{v,i} \epsilon_{w,j}. \quad (3.1)$$

Summing over the reversed pairs gives $K(\epsilon + \epsilon') - K(\epsilon) - K(\epsilon')$. The local definitions (2.10) and (2.9) supply the remaining $\sum_v \epsilon_{v,2} \epsilon'_{v,3}$. Thus the required convolution is

$$\left(\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right) \star \left(\prod_e q_e^{s_e} \eta_e^{\epsilon'_e} \right) = (-1)^{K(\epsilon + \epsilon') - K(\epsilon) - K(\epsilon') + \sum_v \epsilon_{v,2} \epsilon'_{v,3}} \prod_e q_e^{r_e + s_e} \eta_e^{\epsilon_e + \epsilon'_e}. \quad (3.2)$$

Internal edge parities determine the corresponding vertex-slot parities. External parities are fixed labels of the two inputs; the result carries their sum. The sign is bilinear in the two parity assignments.

This proves associativity by expanding either parenthesization of a product of three monomials. Keep the physical factor on the left, since the product need not be commutative. The η_e are formal parity labels with ordinary relation $\eta_e^2 = 1$; a convolution square may carry an additional minus sign from (3.2). Evaluate the final physical block at $\eta_e = \pm 1$ after the coefficient calculation. Substitution before convolution would lose these signs.

Expanding the tensor-PBW sewing gives

$$\widehat{\mathbf{F}}_{\hat{\Gamma}} = \mathbf{F}_{\hat{\Gamma}} \star \mathbf{F}_{\mathbf{F}, \hat{\Gamma}}. \quad (3.3)$$

Every sign comes from a pairing, a local exchange, or the global permutation already specified. The physical factor is precisely the block defined in (1.7).

We can now write the relation coefficient by coefficient, as in the theta-channel derivation in the SCblock notes. Denote the coefficient of $\prod_e q_e^{n_e} \eta_e^{\delta_e}$ by $\widehat{\mathbf{F}}_{\hat{\Gamma}}[\prod_e q_e^{n_e} \eta_e^{\delta_e}]$, where n_e is the enlarged level and $\delta_e \in \{0, 1\}$ is its parity on edge e . Each contribution splits this level as $n_e = r_e + s_e$, with r_e from the physical block and s_e from the fermion block. Their parities satisfy $\epsilon_e + \epsilon'_e = \delta_e$ modulo two. Substituting (3.2), we obtain

$$\begin{aligned} \widehat{\mathbf{F}}_{\hat{\Gamma}} \left[\prod_e q_e^{n_e} \eta_e^{\delta_e} \right] &= \sum_{\substack{r_e + s_e = n_e \text{ for every } e \\ \epsilon_e + \epsilon'_e = \delta_e \pmod{2}}} (-1)^{K(\delta) - K(\epsilon) - K(\epsilon')} \prod_v (-1)^{\epsilon_v, 2\epsilon'_{v,3}} \\ &\quad \times \mathbf{F}_{\hat{\Gamma}} \left[\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right] \mathbf{F}_{\mathbf{F}, \hat{\Gamma}} \left[\prod_e q_e^{s_e} \eta_e^{\epsilon'_e} \right]. \end{aligned} \quad (3.4)$$

The sum runs over the allowed physical and auxiliary levels and parities on every internal edge; the vertex labels and external states remain fixed. Components violating (1.5) or auxiliary parity conservation are zero. In $K(\delta)$, the external parities are also the sums of those of the two factors. Thus each summand is a physical coefficient times a fermion coefficient, multiplied by the global reordering sign and the local mixed signs. All brackets in (3.4) select scalar coefficients. Brackets containing only $\prod_e q_e^{n_e}$ instead retain the polynomial in the spin lifts.

For example, in the theta channel the two vertices have the same slots and equal a or f labels. Their local mixed signs multiply to $(-1)^{2\epsilon_2\epsilon'_3} = 1$. The only remaining sign is

$$(-1)^{K(\delta) - K(\epsilon) - K(\epsilon')} = (-1)^{\sum_{i < j} (\epsilon_i \epsilon'_j + \epsilon'_i \epsilon_j)}, \quad \delta_i = \epsilon_i + \epsilon'_i \pmod{2},$$

which gives the theta-channel convolution of the SCblock notes. In particular, $\widehat{\mathbf{F}}_{\theta} = \mathbf{F}_{\theta} \star \mathbf{F}_{\mathbf{F}, \theta}$. On a general graph the local factors in (3.4) need not cancel; they are included in $\star$. Recovery solves directly for $\mathbf{F}_{\hat{\Gamma}}$.

4 Reduction to double Virasoro

Each enlarged internal module is a direct sum of modules for the two commuting Virasoro algebras. Their primaries are the v_n and v_n^α of the draft. The label ranges and parameters are

$$\begin{aligned} n &\in \tfrac{1}{2}\mathbb{Z} \quad (\text{NS}), & n &\in \tfrac{1}{2}\mathbb{Z} + \tfrac{1}{4} \quad (\text{R}), & \alpha &= 0, 1, \\ Q &= b + b^{-1}, & c &= \tfrac{3}{2} + 3Q^2, & b &\neq 0, \pm 1, \\ c^{(1)} &= 1 + \frac{3Q^2}{1 - b^2}, & c^{(2)} &= 1 + \frac{3Q^2}{1 - b^{-2}}, \\ h_{e,n}^{(1)} &= \frac{Q^2/4 - (P_e + 2nb)^2}{2(1 - b^2)}, & h_{e,n}^{(2)} &= \frac{Q^2/4 - (P_e + 2n/b)^2}{2(1 - b^{-2})}. \end{aligned} \quad (4.1)$$

For an NS module $P_e^2 = Q^2/4 - 2h_e$; for a Ramond module $P_e^2 = 2\beta_e^2$ and $h_e = c/24 - \beta_e^2$. The label on v_n^0, v_n^1 is the total even or odd parity of an enlarged primary, which mixes auxiliary and physical Ramond grounds. The states $w^\pm$ used in the PBW sum are the physical grounds before this change of basis.

Subtracting the enlarged ground weight gives the starting level of each branch. The NS auxiliary ground has weight zero, and the Ramond auxiliary ground has weight $1/16$. Therefore

	branch starting level	total parity ϵ_e
NS	$h_{e,n}^{(1)} + h_{e,n}^{(2)} - h_e = 2n^2$	$(2n) \bmod 2$
R	$h_{e,n}^{(1)} + h_{e,n}^{(2)} - h_e - \frac{1}{16} = 2n^2 - \frac{1}{8}$	α_e

(4.2)

Thus ϵ in this section is the parity of the sewn branch state; in the PBW convolution it was the parity of the physical factor. All states within a fixed double-Virasoro module have the same parity.

The two types of normalized branching coefficient are

$$\begin{aligned} \mathbf{B}_a(n_1, n_2, n_3) &= \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \\ \mathbb{B}_f^{(\eta)}(n_1, n_2, n_3; \alpha_2, \alpha_3) &= \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}. \end{aligned} \quad (4.3)$$

Here $\|v\|^2 = \langle\langle v|v \rangle\rangle$ is a bilinear BPZ norm, not an absolute square. Choose its roots consistently for all vertices and edges. We suppress only the fixed representation parameters in $\mathbf{B}, \mathbb{B}$; the branch labels and Ramond parities are displayed.

First take normalized branch primaries at the external punctures. The enlarged block is then

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{\Gamma}} &= \sum_{\{n_e, \alpha_e\}} (-1)^{K(\epsilon)} \prod_{e:\text{NS}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{e:\text{R}} q_e^{2n_e^2 - 1/8} \eta_e^{\alpha_e} \\ &\times \prod_{v:\text{NS,NS,NS}} \mathbf{B}_{a_v}(n_{v,1}, n_{v,2}, n_{v,3}) \\ &\times \prod_{v:\text{NS,R,R}} \mathbb{B}_{f_v}^{(\eta_v)}(n_{v,1}, n_{v,2}, n_{v,3}; \alpha_{v,2}, \alpha_{v,3}) \prod_{j=1}^2 F_{\hat{\Gamma}}(\{h_{e,n_e}^{(j)}\}; c^{(j)}). \end{aligned} \quad (4.4)$$

The sum and edge products run over internal edges; external branch labels are fixed. The argument list of each Virasoro block includes both the internal weights and the fixed external weights. A slot label $n_{v,i}$ is the label of its incident edge, so a self-loop has one branching sum shared by its two slots. Keep only the local parity components allowed by (1.5), now applied to total branch parities.

To see the reduction, change basis separately on every internal edge. The BPZ form becomes the product of the two Virasoro Gram matrices and the primary norm. The two adjacent normalized branching coefficients supply the inverse primary norm once per edge. The Virasoro Ward identities then reduce descendants independently in the two copies. For example, at an NS–R–R vertex,

$$\begin{aligned} \hat{\rho}_f^{(\eta)}(L_{-1}^{(j)} v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\ = (h_{1,n_1}^{(j)} + h_{2,n_2}^{(j)} - h_{3,n_3}^{(j)}) \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}), \quad j = 1, 2. \end{aligned} \quad (4.5)$$

Both copies obey this identity with (2.10); the directed check uses the explicit oscillator embeddings. An external tensor-PBW state is handled by the finite expansion in Sec. 8, followed by the same Ward reduction.

In computation, evaluate the Virasoro factors by CCY c -recursion. Its large- c seed contains the L_{-1} global block and the universal Schottky vacuum factor. Store pole geometry, fusion polynomials,

residue normalizations, global vertices, and repeated recursion states. At NS-R-R vertices obtain the outer branching coefficients recursively from $n_{\text{NS}} = 0, \pm\frac{1}{2}$ and $n_{\text{R}} = \pm\frac{1}{4}, \pm\frac{3}{4}$, reusing intermediate coefficients and the coefficients of the $L_{\pm 1}$ descendant expansions. When a Ward equation has a zero leading coefficient, use the alternative with L_1 at a Ramond slot. The validation driver obtains its all-NS coefficients from free-field primary vectors and caches those local values. The remaining Virasoro cutoff depends on the branch: subtract its starting level in (4.2) before evaluating descendants. This prevents computing blocks beyond the requested physical degree.

5 Ramond loops, cancellation, and the constant term

Every vertex meets zero or two Ramond edges. The Ramond subgraph therefore consists of disjoint closed cycles and paths with external Ramond ends. A self-loop is a closed cycle with one vertex. Let C be any closed Ramond cycle. We write $K(C)$ for the already defined inversion count K , evaluated with parity one on that cycle and zero on all other internal and external punctures. No new sign rule is involved.

The physical coefficients on the cycle are related by

$$\left(\prod_{e \in C} \eta_e\right) \star \mathbf{F}_{\hat{\Gamma}} = (-1)^{K(C) + \sum_{v \in C} f_v} \left(\prod_{v \in C} i\eta_v\right) \mathbf{F}_{\hat{\Gamma}}. \quad (5.1)$$

The left side flips the retained parity on every edge of C , with the convolution signs of (3.2). The right side fixes the relative coefficients of those components. It is a sector identity for the formal series; it does not assign complex values to the spin lifts.

Locally this identity follows from the odd ground map $w^+ \mapsto e^{-i\pi/4}w^-$ and $w^- \mapsto -e^{i\pi/4}w^+$, extended to a Ramond PBW word with the factor $(-1)^A$. This map preserves the ground BPZ form $\text{diag}(1, i)$. Acting on both Ramond slots multiplies the physical form by $-i\eta_v(-1)^{f_v + \epsilon_{v,2}}$; the ground values (1.4) verify it, and the supercurrent Ward identities extend it to descendants. Combining these local identities around the cycle and reordering the odd maps gives (5.1). Crossing the NS slot twice contributes $(-1)^{2\epsilon_{v,1}} = 1$.

The cycle monomial commutes with admissible factors of the convolution: its commutator exponent is

$$\sum_{v \in C} (\epsilon_{v,2} + \epsilon_{v,3}) = 2 \sum_{e \in C} \epsilon_e = 0 \pmod{2}. \quad (5.2)$$

Each Ramond edge is counted once at each end. The global terms from K cancel in the commutator because they are symmetric in its two inputs. Applying the same ground-map argument to the auxiliary form gives

$$\left(\prod_{e \in C} \eta_e\right) \star \mathbf{F}_{\mathbf{F}, \hat{\Gamma}} = (-1)^{K(C)} \left(\prod_{v \in C} i\right) \mathbf{F}_{\mathbf{F}, \hat{\Gamma}}. \quad (5.3)$$

Move this commuting monomial between the two factors of (3.4) and cancel the common phase. In one line,

$$\left[(-1)^{\sum_{v \in C} f_v} \prod_{v \in C} \eta_v - 1 \right] \hat{\mathbf{F}}_{\hat{\Gamma}} = 0. \quad (5.4)$$

Thus the enlarged block vanishes whenever

$$(-1)^{\sum_{v \in C} (f_v)} \prod_{v \in C} \eta_v = -1. \quad (5.5)$$

Its two factors belong to incompatible cycle sectors; the physical block itself can be nonzero. In theta the two equal f labels cancel, leaving $\eta\eta' = -1$.

The auxiliary constant term makes both the obstruction and the restricted inverse concrete. With the external ground states chosen in Sec. 2, all NS auxiliary slots are vacuum at level zero. Around a closed Ramond cycle the auxiliary grounds must then be either all u^0 or all u^1 . The first choice contributes one. The second contributes i from each vertex, -1 from each inverse ground metric, its spin lifts, and the ordering sign. Consequently,

$$\mathbf{F}_{\mathbb{F}, \hat{\Gamma}} \Big|_{q_e=0} = \prod_C^* \left[1 + (-1)^{K(C)} \left(\prod_{v \in C} (-i) \right) \left(\prod_{e \in C} \eta_e \right) \right]. \quad (5.6)$$

The product runs over the closed Ramond cycles and is taken using $\star$; its factors commute. The empty product is one, and open paths with u^0 ends contribute one. Each term after the 1 in a bracket has convolution square one: the cycle monomial has square $(-1)^{\text{number of vertices on } C}$, canceled by the squared factors $-i$. Hence each bracket acts by two on its matching sector and by zero on the other sector. The constant is invertible on the matching simultaneous sector, but not on the full coefficient space if a closed Ramond cycle is present.

6 Splitting an obstructed edge and recovering the block

6.1 The inserted sphere and its PBW matrix element

Choose one edge on each cycle satisfying (5.5). On a chosen edge e , insert a new sphere with coordinate w , sew its punctures $0, \infty$ to the original ends, and place $\Theta\psi$ at $w = 1$. If z_L, z_R are the original local coordinates, the two sewing equations are $z_L w = q_{e_L}$ and $z_R/w = q_{e_R}$. Multiplying them gives

$$q_e = q_{e_L} q_{e_R}, \quad \eta_e = \eta_{e_L} \eta_{e_R}. \quad (6.1)$$

The spin lifts multiply in the aligned Ramond frames, with no extra odd insertion sign because $\Theta\psi$ is even. The operator is defined by

$$\Theta u^a = (-1)^{a+1} u^{1-a}, \quad \{\Theta, \psi_r\} = \{\Theta, G_r\} = 0, \quad [\Theta, L_n] = 0. \quad (6.2)$$

Both factors anticommute with physical G_r , so their product commutes with the physical SCA. Also Θ commutes with $L^{(1,2)}$: it commutes with the even fermion bilinears and with ψG . The inserted vertex is the radial matrix element of this operator.

The physical descendant words at the inserted sphere are B_L, B_R . Their auxiliary words are $\mathbb{B}_L, \mathbb{B}_R$, with ground labels $\mathbf{b}_L, \mathbf{b}_R$. The matrix element is

$$\begin{aligned} & \left\langle \left\langle \Psi_{-\mathbb{B}_R} u^{\mathbf{b}_R} \otimes \mathbb{L}_{-B_R} w^{\alpha_R} \middle| \Theta\psi(1) \middle| \Psi_{-\mathbb{B}_L} u^{\mathbf{b}_L} \otimes \mathbb{L}_{-B_L} w^{\alpha_L} \right\rangle \right. \\ & \quad \left. = \langle \Psi_{-\mathbb{B}_R} u^{\mathbf{b}_R} | \Theta\psi(1) | \Psi_{-\mathbb{B}_L} u^{\mathbf{b}_L} \rangle (\mathbb{G}_{\beta_e})_{B_R \alpha_R, B_L \alpha_L}. \end{aligned} \quad (6.3)$$

The order fixes Θ next to the outgoing bra. Each new segment has its own physical inverse Gram matrix, auxiliary inverse metric, and factor $q^{|B|+|\mathbb{B}|} \eta^\epsilon (-\eta)^{\epsilon'}$ as in (2.11).

6.2 Definition of the enlarged block with split edges

Keep the original labels $A_{e,1}, \alpha_{e,1}$ and $A_{e,2}, \alpha_{e,2}$ at the two outer ends of a split Ramond edge e . The physical states on the inserted sphere have labels B_{e_L}, α_{e_L} at its ket puncture 0 and B_{e_R}, α_{e_R} at its bra puncture ∞ . The auxiliary inverse Gram matrices are diagonal: the word $\mathbb{B}_{e_L}$ and ground

label $\mathbf{b}_{e_L}$ are shared by the two ends of segment e_L , and similarly for e_R . Thus $\epsilon'_{e_L} = \mathbf{B}_{e_L} + \mathbf{b}_{e_L}$ and $\epsilon'_{e_R} = \mathbf{B}_{e_R} + \mathbf{b}_{e_R}$ modulo two.

The inserted matrix element (6.3) identifies the physical levels and parities across the split edge, and preserves auxiliary parity. On every nonzero term, $\epsilon_{e_L} = \epsilon_{e_R} = \epsilon_e$ and $\epsilon'_{e_L} = \epsilon'_{e_R} = \epsilon'_e$. It does not identify the auxiliary levels. With independent plumbing parameters on all segments, the full enlarged block is

$$\begin{aligned}
\hat{\mathbf{F}}_{\hat{\Gamma}}[\Theta\psi] = & \sum_{\{A,\alpha,B,\mathbf{A},\mathbf{a},\mathbf{B},\mathbf{b}\}} (-1)^{K(\epsilon+\epsilon')} \prod_{e \text{ unsplit}} q_e^{|A_{e,1}|+|A_e|} \eta_e^{\epsilon_e+\epsilon'_e} (-1)^{\epsilon'_e} \\
& \times \prod_{e:\text{NS}} \mathbf{G}_{h_e}^{A_{e,1}A_{e,2}} \prod_{\substack{e:\text{R} \\ e \text{ unsplit}}} \mathbb{G}_{\beta_e}^{A_{e,1}\alpha_{e,1},A_{e,2}\alpha_{e,2}} \\
& \times \prod_{e \text{ split}} \left[q_{e_L}^{|B_{e_L}|+|\mathbf{B}_{e_L}|} q_{e_R}^{|B_{e_R}|+|\mathbf{B}_{e_R}|} \eta_{e_L}^{\epsilon_e+\epsilon'_{e_L}} \eta_{e_R}^{\epsilon_e+\epsilon'_{e_R}} (-1)^{\epsilon'_{e_L}+\epsilon'_{e_R}} \right. \\
& \quad \times \mathbb{G}_{\beta_e}^{A_{e,1}\alpha_{e,1},B_{e_L}\alpha_{e_L}} \mathbb{G}_{\beta_e}^{B_{e_R}\alpha_{e_R},A_{e,2}\alpha_{e,2}} \\
& \quad \times \left\langle \left\langle \Psi_{-\mathbf{B}_{e_R}} u^{\mathbf{b}_{e_R}} \otimes \mathbb{L}_{-B_{e_R}} w^{\alpha_{e_R}} \middle| \Theta\psi(1) \middle| \Psi_{-\mathbf{B}_{e_L}} u^{\mathbf{b}_{e_L}} \otimes \mathbb{L}_{-B_{e_L}} w^{\alpha_{e_L}} \right\rangle \right\rangle \\
& \times \prod_{v:\text{NS},\text{NS},\text{NS}} \hat{\rho}_{a_v}(\Psi_{-A_{v,1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,1}} \phi_{v,1}, \\
& \quad \Psi_{-A_{v,2}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,2}} \phi_{v,2}, \Psi_{-A_{v,3}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,3}} \phi_{v,3}) \\
& \times \prod_{v:\text{NS},\text{R},\text{R}} \hat{\rho}_{f_v}^{(\eta_v)}(\Psi_{-A_{v,1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v,1}} \phi_{v,1}, \\
& \quad \Psi_{-A_{v,2}} u^{\mathbf{a}_{v,2}} \otimes \mathbb{L}_{-A_{v,2}} w^{\alpha_{v,2}}, \Psi_{-A_{v,3}} u^{\mathbf{a}_{v,3}} \otimes \mathbb{L}_{-A_{v,3}} w^{\alpha_{v,3}}). \tag{6.4}
\end{aligned}$$

The sum includes all displayed internal words and Ramond ground labels; physical ground labels take values $+$, $-$, and auxiliary ground labels take values $0, 1$. The two physical inverse Gram matrices on a split edge impose $|A_{e,1}| = |B_{e_L}|$ and $|B_{e_R}| = |A_{e,2}|$. The vertex products contain only the original spheres. At their slots, the auxiliary labels $A_{v,i}, \mathbf{a}_{v,i}$ are those of the adjacent segment: $\mathbf{B}_{e_L}, \mathbf{b}_{e_L}$ at the original end $(e, 1)$ and $\mathbf{B}_{e_R}, \mathbf{b}_{e_R}$ at $(e, 2)$. Unsplit-edge labels and fixed external states are as in (2.11). Each new sphere appears exactly once, through the displayed matrix element.

We keep the original edge and vertex order and insert the even matrix element in its oriented edge contraction. Hence K is the original graph's reordering exponent, evaluated with the common segment parity. The two factors $(-1)^{\epsilon'_{e_L}}$ and $(-1)^{\epsilon'_{e_R}}$ are the two auxiliary inverse Gram entries; the BPZ pairing inside the matrix element is also retained. For theta, with edge 2 split, the product of all auxiliary inverse Gram signs is $(-1)^{\mathbf{B}_R+\mathbf{b}_R}$ by outer-vertex parity conservation. This is an edge factor, not a phase of either three-point form.

Equation (6.4) is the full split series, before the diagonal restriction: q_{e_L} and q_{e_R} remain independent, and $|\mathbf{B}_{e_L}|$ and $|\mathbf{B}_{e_R}|$ may differ. It applies to any collection of split Ramond edges, including several disjoint loops. When no edge is split it reduces to (2.11).

Substituting (6.3) into this definition and using the symmetry of the physical BPZ Gram matrix reduces the physical contraction to

$$\begin{aligned}
& \sum_{B_{e_L}, B_{e_R}, \alpha_{e_L}, \alpha_{e_R}} \mathbb{G}_{\beta_e}^{A_{e,1}\alpha_{e,1}, B_{e_L}\alpha_{e_L}} (\mathbb{G}_{\beta_e})_{B_{e_L}\alpha_{e_L}, B_{e_R}\alpha_{e_R}} \mathbb{G}_{\beta_e}^{B_{e_R}\alpha_{e_R}, A_{e,2}\alpha_{e,2}} \\
& = \mathbb{G}_{\beta_e}^{A_{e,1}\alpha_{e,1}, A_{e,2}\alpha_{e,2}}. \tag{6.5}
\end{aligned}$$

This contracts the identity operator on the physical line. Its levels and parities agree on the two segments. Auxiliary parities agree because $\Theta\psi$ is even, but auxiliary levels may differ through the nonzero modes of $\psi(1)$. Using only the auxiliary matrix element in (6.3) defines the inserted fermion block by the same sewing formula: omit the physical words, ground labels, and their sums, the physical Gram factors, and the physical descendant powers of q ; replace each outer form by ρ_F on its auxiliary arguments and set every physical ϵ to zero. The resulting expression is $\mathbf{F}_{\hat{F}}[\Theta\psi]$, with reordering sign $(-1)^{K(\epsilon')}$ and purely auxiliary edge levels and parities.

6.3 The inserted double-Virasoro sum

The field ψ is proportional to the NS branch primary $v_{1/2}$ at $P = Q/2$. Its two Virasoro weights are

$$h_\psi^{(1)} = -\frac{1+2b^2}{2(1-b^2)}, \quad h_\psi^{(2)} = \frac{b^2+2}{2(1-b^2)}, \quad h_\psi^{(1)} + h_\psi^{(2)} = \frac{1}{2}. \quad (6.6)$$

Their degenerate fusion rules give $n_{e_R} = n_{e_L} \pm \frac{1}{2}$. The odd ψ reverses the total Ramond parity and Θ reverses it back, so the two segments carry the same α_e . The normalized middle coefficient is

$$\mathbb{B}(n_{e_R}, n_{e_L}, \alpha_e) = \frac{\langle v_{e,n_{e_R}}^{\alpha_e} | \Theta\psi(1) | v_{e,n_{e_L}}^{\alpha_e} \rangle}{\|v_{e,n_{e_R}}^{\alpha_e}\| \|v_{e,n_{e_L}}^{\alpha_e}\|}. \quad (6.7)$$

It is zero outside the stated fusion rule. Since Θ commutes with both Virasoro copies, the descendant dependence is entirely in their ordinary punctured blocks.

With normalized branch primaries at the original external punctures, the full sum is

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{F}}[\Theta\psi] &= \sum_{\{n,\alpha\}} (-1)^{K(\epsilon)} \prod_{e:\text{NS}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{\substack{e:\text{R} \\ e \text{ unsplit}}} q_e^{2n_e^2-1/8} \eta_e^{\alpha_e} \\ &\times \prod_{e \text{ split}} q_{e_L}^{2n_{e_L}^2-1/8} q_{e_R}^{2n_{e_R}^2-1/8} (\eta_{e_L} \eta_{e_R})^{\alpha_e} \mathbb{B}(n_{e_R}, n_{e_L}, \alpha_e) \\ &\times \prod_{v:\text{NS,NS,NS}} \mathbf{B}_{a_v}(n_{v,1}, n_{v,2}, n_{v,3}) \\ &\times \prod_{v:\text{NS,R,R}} \mathbb{B}_{f_v}^{(\eta_v)}(n_{v,1}, n_{v,2}, n_{v,3}; \alpha_{v,2}, \alpha_{v,3}) \\ &\times \prod_{j=1}^2 F_{\hat{F}}[\psi](\{h_{e,n_e}^{(j)}\}; \{h_\psi^{(j)}\}; c^{(j)}). \end{aligned} \quad (6.8)$$

Vertex products here include only the original spheres; the middle coefficients account for the new spheres once each. The internal branch sum includes both labels n_{e_L}, n_{e_R} on a split edge, constrained by the fusion rule, and one common parity α_e . In an outer branching coefficient, $n_{v,i}$ is the label of the segment adjacent to that original vertex. The function $F_{\hat{F}}[\psi]$ denotes the ordinary Virasoro block on this expanded channel, with those segment weights, the original external weights, and one additional external weight $h_\psi^{(j)}$ for each inserted sphere. It is computed by the same CCY recursion. The original graph's K applies because the inserted operator is even and the segment parities agree. Outer and middle coefficients use the same BPZ norm roots.

6.4 Convolution before and after the diagonal restriction

Equation (6.5) shows that the full physical monomial is replaced by

$$\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \longmapsto \prod_{e \text{ unsplit}} q_e^{r_e} \eta_e^{\epsilon_e} \prod_{e \text{ split}} (q_{e_L} q_{e_R})^{r_e} (\eta_{e_L} \eta_{e_R})^{\epsilon_e}. \quad (6.9)$$

Thus the complete unprojected convolution is (3.4) with $\widehat{\mathbf{F}}_{\hat{\Gamma}}$ and $\mathbf{F}_{\mathbf{F},\hat{\Gamma}}$ replaced by their inserted blocks, and every physical monomial replaced by (6.9). The physical coefficient itself is still that of the original graph. This is an exact substitution in the displayed formula, including its mixed sign. In particular, a physical power r_e and auxiliary powers (s_{e_L}, s_{e_R}) produce $(r_e + s_{e_L}, r_e + s_{e_R})$; unequal split powers remain included.

To recover the physical block it suffices to keep the terms with equal powers on each split pair. Define the subscript diag precisely by coefficient extraction:

$$\begin{aligned} \widehat{\mathbf{F}}_{\hat{\Gamma}}[\Theta\psi]_{\text{diag}} & \left[\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right] \\ &= \widehat{\mathbf{F}}_{\hat{\Gamma}}[\Theta\psi] \left[\prod_{e \text{ unsplit}} q_e^{r_e} \eta_e^{\epsilon_e} \prod_{e \text{ split}} (q_{e_L} q_{e_R})^{r_e} (\eta_{e_L} \eta_{e_R})^{\epsilon_e} \right]. \end{aligned} \quad (6.10)$$

Use the identical definition for the fermion factor. Since the physical powers on a split pair already agree, equality of the final powers is equivalent to equality of the auxiliary powers. Hence the diagonal restriction commutes with the convolution.

On the fermion side this keeps $\Theta\psi_0$. Its action is particularly simple:

$$\Theta\psi_0 \Psi_{-A} u^a = \Psi_{-A} \Theta\psi_0 u^a = \frac{(-1)^a}{\sqrt{2}} \Psi_{-A} u^a. \quad (6.11)$$

Moving each of Θ and ψ_0 through the nonzero fermions gives $(-1)^A$, so those two signs cancel. Only the ground label remains. The ground factor on the affected cycle changes from the bracket in (5.6) to

$$\frac{1}{\sqrt{2}} \left[1 - (-1)^{K(C)} \left(\prod_{v \in C} (-i) \right) \left(\prod_{e \in C} \eta_e \right) \right]. \quad (6.12)$$

It now selects the previously missing sector. Each closed cycle acts by two on its selected sector, with an additional $1/\sqrt{2}$ for each insertion. No inverse on the unrestricted coefficient space is asserted.

The Virasoro calculation can enforce the same restriction before summing:

$$2n_{e_L}^2 - \frac{1}{8} + k_{e_L}^{(1)} + k_{e_L}^{(2)} = 2n_{e_R}^2 - \frac{1}{8} + k_{e_R}^{(1)} + k_{e_R}^{(2)}. \quad (6.13)$$

Here $k^{(1)}, k^{(2)}$ are the nonnegative integer descendant levels in the two copies. The restriction concerns their combined power together with the branch starting level. It does not set $n_{e_L} = n_{e_R}$ or match the descendant levels separately in each copy. Use it together with the physical cutoff to discard unneeded branch tuples and recursion states, and reuse the middle and outer branching coefficients.

6.5 The coefficient recursion

After inserting once on every obstructed cycle, the diagonal auxiliary constant is nonzero on the required sector. The following coefficient equation makes the triangular solve explicit. The level indices r_e, s_e, t_e are nonnegative half-integers on NS edges and integers on Ramond edges; coefficients absent from the corresponding series are zero. Here $\prod_C 2$ runs over the original closed Ramond

cycles, and the other constant factor runs over the edges split by insertions:

$$\begin{aligned}
& \widehat{\mathbf{F}}_{\hat{\Gamma}}[\Theta\psi]_{\text{diag}} \left[\prod_e q_e^{r_e} \right] \\
&= \sum_{\epsilon} \left\{ \left(\prod_C 2 \right) \left(\prod_{e \text{ split}} \frac{1}{\sqrt{2}} \right) \mathbf{F}_{\hat{\Gamma}} \left[\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right] \prod_e \eta_e^{\epsilon_e} \right. \\
&\quad \left. + \sum_{\substack{s_e+t_e=r_e \text{ for all } e \\ s_e, t_e \geq 0, \sum_e t_e > 0}} \mathbf{F}_{\hat{\Gamma}} \left[\prod_e q_e^{s_e} \eta_e^{\epsilon_e} \right] \left(\prod_e \eta_e^{\epsilon_e} \right) \star \mathbf{F}_{\mathbf{F}, \hat{\Gamma}}[\Theta\psi]_{\text{diag}} \left[\prod_e q_e^{t_e} \right] \right\}. \tag{6.14}
\end{aligned}$$

This is an equality of parity polynomials. The second sum uses only lower total physical levels because $\sum_e t_e > 0$. Subtract it from the known enlarged coefficient and divide by the displayed constant to obtain the desired physical coefficient. At level zero the second sum is empty. For an unobstructed calculation omit the insertion brackets, the diagonal restriction, and the product over split edges.

The equations above use the canonical fermion normalization (2.2). The implementation inserts $Q\Theta\psi$ in both the enlarged numerator and the auxiliary factor. For that convention, replace each $1/\sqrt{2}$ in the constant by $Q/\sqrt{2}$ and multiply each middle coefficient by Q . The factors cancel from the recovered block. At no stage is the physical PBW result an input to this recursion; it enters only as an independent reference in the validation section.

7 Theta, glasses, and Mercedes

7.1 Theta

Use the two orders given after (1.6). The resulting signs are

$$K(\epsilon) = \epsilon_1\epsilon_2 + \epsilon_1\epsilon_3 + \epsilon_2\epsilon_3, \quad (-1)^{\sum_{i < j} (\epsilon_i\epsilon'_j + \epsilon'_i\epsilon_j)} \quad \text{in the convolution.} \tag{7.1}$$

The two local mixed terms in (3.2) cancel. The cycle identity and auxiliary ground factors become

$$\begin{aligned}
& (\eta_2\eta_3) \star \mathbf{F}_{\theta} = \eta\eta' \mathbf{F}_{\theta}, \\
& \mathbf{F}_{\mathbf{F}, \theta}|_{q=0} = 1 + \eta_2\eta_3, \quad \mathbf{F}_{\mathbf{F}, \theta}[\Theta\psi]_{\text{diag}}|_{q=0} = \frac{1 - \eta_2\eta_3}{\sqrt{2}}. \tag{7.2}
\end{aligned}$$

The ordinary factor selects $\eta\eta' = 1$; the inserted factor selects $\eta\eta' = -1$. Splitting either Ramond edge supplies the missing sector. All formulas use the local forms and BPZ pairings defined above.

7.2 Glasses

Use edge 1 as the NS bridge and edges 2,3 as Ramond self-loops, with vertex triples $(1_L, 2_L, 2_R)$ and $(1_R, 3_L, 3_R)$. Take an even bridge primary. Parity conservation fixes the same f at both vertices. Now

$$K(\epsilon) = 0, \quad (-1)^{\epsilon_2\epsilon'_2 + \epsilon_3\epsilon'_3} \quad \text{in the convolution.} \tag{7.3}$$

In particular $\eta_2 \star \eta_2 = \eta_3 \star \eta_3 = -1$. The physical cycle identities are

$$\eta_2 \star \mathbf{F}_{\text{gl}} = (-1)^f i\eta_{v_1} \mathbf{F}_{\text{gl}}, \quad \eta_3 \star \mathbf{F}_{\text{gl}} = (-1)^f i\eta_{v_2} \mathbf{F}_{\text{gl}}. \tag{7.4}$$

The ordinary auxiliary constant is $(1 - i\eta_2)(1 - i\eta_3)$. Inserting on edge 2 replaces its factor by $(1 + i\eta_2)/\sqrt{2}$, and similarly for edge 3. The loop at vertex v needs an insertion when $(-1)^f \eta_v = -1$. Thus for $f = 1$ the obstructed sign is $\eta_v = +1$. There can be zero, one, or two inserted spheres; each loop is treated independently. Recovery returns the physical block with the specified vertex labels.

7.3 Mercedes: one external insertion

The genus-three Mercedes channel has a triangular rim and three spokes meeting at a central vertex. Its four vertices and six internal edges give $g = 6 - 4 + 1 = 3$. Take spokes 1, 2, 3 to be NS and rim 4, 5, 6 to be Ramond, with slots

vertex	$(\infty, 1, 0)$ edges	form
center	(1, 2, 3)	ρ_a
v_1	(1, 4, 6)	$\rho_{f_1}^{(\eta_{v_1})}$
v_2	(2, 5, 4)	$\rho_{f_2}^{(\eta_{v_2})}$
v_3	(3, 6, 5)	$\rho_{f_3}^{(\eta_{v_3})}$

(7.5)

Order edges by number, orient each spoke from the center outwards, and orient the rim $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_1$. Take even intrinsic primary parities, as in the saved Mercedes tests. Summing the local parity constraints gives $a = f_1 + f_2 + f_3$ modulo two. There is one Ramond cycle, (4, 5, 6), for which $K(C) = 0$. Its physical sector identity and obstruction are

$$\begin{aligned}
 (\eta_4 \eta_5 \eta_6) \star \mathbf{F}_{\hat{\Gamma}} &= (-1)^{f_1+f_2+f_3} (-i) \eta_{v_1} \eta_{v_2} \eta_{v_3} \mathbf{F}_{\hat{\Gamma}}, \\
 (-1)^{f_1+f_2+f_3} \eta_{v_1} \eta_{v_2} \eta_{v_3} &= -1 \quad \text{for an obstructed loop.}
 \end{aligned}
 \tag{7.6}$$

The edge spin lifts and the three-point-form signs remain distinct. For the ordinary and inserted calculations the auxiliary constants are

$$\mathbf{F}_{\mathbf{F}, \hat{\Gamma}} \Big|_{q=0} = 1 + i\eta_4 \eta_5 \eta_6, \quad \mathbf{F}_{\mathbf{F}, \hat{\Gamma}} [\Theta\psi]_{\text{diag}} \Big|_{q=0} = \frac{1 - i\eta_4 \eta_5 \eta_6}{\sqrt{2}}.
 \tag{7.7}$$

This displays the switch of sector using a single insertion.

If the obstruction holds, split edge 4 with its original ket end at v_1 and bra end at v_2 , and place one external $\Theta\psi$ on the new sphere. All three NS spokes remain sewn to the central sphere. The enlarged calculation is a genus-three one-point block; its convolution recovers the original physical block without an external insertion. There are no additional external NS legs at the three outer vertices. If the loop is unobstructed, use the original four spheres without the insertion. These are the two Mercedes constructions tested below.

8 External states and the full correlator

8.1 Projecting an external state onto the branches

An external tensor-PBW state need not be a branch primary. Expand it in the same orthogonal double-Virasoro basis used on the internal edges. At a Ramond puncture, its physical and auxiliary parities are $\epsilon = A + |\alpha|$ and $\epsilon' = A + \mathbf{a}$ modulo two. Only branch primaries of total parity $\epsilon + \epsilon'$ can occur. Using $G_h^{BB'}$ for the inverse Virasoro Gram matrix at weight h , the BPZ expansion is

$$\begin{aligned}
 &\Psi_{-A} u^{\mathbf{a}} \otimes \mathbb{L}_{-A} w^{\alpha} \\
 &= \sum_{n, B, C, B', C'} (G_{h_n^{(1)}})^{BB'} (G_{h_n^{(2)}})^{CC'} \left\langle \left\langle \frac{L_{-B'}^{(1)} L_{-C'}^{(2)} v_n^{\epsilon+\epsilon'}}{\|v_n^{\epsilon+\epsilon'}\|} \right| \Psi_{-A} u^{\mathbf{a}} \otimes \mathbb{L}_{-A} w^{\alpha} \right\rangle \right. \\
 &\quad \times \frac{L_{-B}^{(1)} L_{-C}^{(2)} v_n^{\epsilon+\epsilon'}}{\|v_n^{\epsilon+\epsilon'}\|}.
 \end{aligned}
 \tag{8.1}$$

The superscript $\epsilon + \epsilon'$ is reduced to zero or one. The finite sum is restricted by $2n^2 - 1/8 + |B| + |C| = |A| + |A|$ and $|B| = |B'|$, $|C| = |C'|$. This is the usual basis-expansion formula: the overlap supplies a covariant component and the inverse Gram matrices convert it to a basis coefficient. It does not require fitting the state to an unknown linear combination.

For NS, replace the physical and auxiliary Ramond grounds by ϕ and $\mathbf{1}$, replace $\mathbb{L}$ by $\mathbf{L}$, and use v_n with starting level $2n^2$. Its parity $2n$ must equal $\epsilon + \epsilon'$. A ground tensor therefore involves only $n = 0$ in NS or $n = \pm 1/4$ in R, with no Virasoro descendants. Insert the expansion at each external puncture in (4.4) or (6.8). Whenever B or C is nonempty, the corresponding Virasoro factor has that external descendant, which is reduced by its own Ward identities. The primary norm roots must match those in the branching coefficients.

8.2 Combining holomorphic and antiholomorphic blocks

Tildes denote the independent antiholomorphic data. For each choice of local form labels, let C_v be the full three-point structure constant with the three holomorphic operators ordered before the three antiholomorphic operators. The normalized chiral forms and BPZ metrics are those already specified. Write $\mathcal{O}_a$ for the external full fields in their prescribed order; a, b below index these punctures, not the form label a_v . The correlator in the plumbing trivialization is

$$\left\langle \prod_a \mathcal{O}_a \right\rangle = \sum_{\substack{\text{internal representations and} \\ \text{holomorphic/antiholomorphic vertex labels}}} (-1)^{\sum_{v < w} (\sum_i \tilde{\epsilon}_{v,i}) (\sum_j \epsilon_{w,j}) + \sum_{a < b} \tilde{\epsilon}_a \epsilon_b} \left(\prod_v C_v \right) \times \left(\prod_e q_e^{h_e} \tilde{q}_e^{h_e} \right) \mathbf{F}_{\hat{\Gamma}} \tilde{\mathbf{F}}_{\hat{\Gamma}}. \quad (8.2)$$

The first sign moves the antiholomorphic operators at a vertex past the holomorphic operators at later vertices. The second restores the prescribed ordering of the external full fields. Each vertex's total parity is fixed by (1.5), so the sign can be taken outside the descendant sums. For an even full three-point tensor the holomorphic and antiholomorphic total parities agree; forbidden label choices have $C_v = 0$.

One can verify the sign directly before doing the descendant sums. Graded factorization of the inverse pairings and three-point forms gives

$$\begin{aligned} K(\epsilon + \tilde{\epsilon}) - K(\epsilon) - K(\tilde{\epsilon}) + \sum_e \epsilon_e \tilde{\epsilon}_e + \sum_v \sum_{i < j} \tilde{\epsilon}_{v,i} \epsilon_{v,j} \\ = \sum_{v < w} \left(\sum_i \tilde{\epsilon}_{v,i} \right) \left(\sum_j \epsilon_{w,j} \right) + \sum_{a < b} \tilde{\epsilon}_a \epsilon_b \pmod{2}. \end{aligned} \quad (8.3)$$

Each internal edge appears twice with equal parities. Moving these even pairs as units cancels their internal contributions and leaves the vertex and external-operator exchanges on the right. This accounts for the sign in (8.2) without assuming Ramond forms are absolute squares.

Sums over a continuous spectrum become integrals. A nontrivial primary multiplicity metric contributes its inverse to the representation sum; the physical Ramond ground metric is already in $\mathbb{G}$. A choice of surface metric also supplies the common conformal-anomaly factor. The block construction determines the chiral sewing coefficients, while the spectrum and the C_v are additional data of the SCFT.

9 Directed numerical validation

This section records all directed tests supporting these notes, including the complete total-level-six extension in Sec. 9.2. The tests distinguish physical SCA PBW sewing, enlarged PBW sewing, double Virasoro assembly, direct fermion sewing, and physical recovery. The double Virasoro numerators are constructed from branching coefficients and Virasoro blocks, without using the physical PBW answers. Numerical discrepancies are reported as $|x-y|/\max(1, |x|, |y|)$; constrained zero components are included. A vanishing test reports the absolute residual. The first subsection records fresh checks of the production convention specified in this version. Subsequent numerical tables are archived results from September 16: their coefficients used the then-current local normalizations and are not fresh checks of the revised headers. They are retained as a record of the requested higher-level tests; the current conventions are exclusively those defined in Sections 1–8.

9.1 Fresh production-convention checks, September 22

The production physical Ward solver, auxiliary Ward solver, boundary branching values, double-Virasoro assembly, and direct PBW executable use the definitions in Sections 1–3. In particular, the enlarged local form contains only the middle-to-ket sign; its inverse-Gram sewing introduces no extra vertex phase. Branching values are computed afresh and stored within the run. All CCY caching, recursive branching, diagonal cutoffs, and Schottky restoration described above remain in use.

An independent residue solver constructs the two forms from the displayed Ward identities and ground values. It shares module actions, but neither physical nor auxiliary Ward solutions, with the production evaluator. It also builds the physical Gram matrices directly in the stated ground basis. Tests use $b = 7/5$, $(P_1, P_2, P_3) = (11/23, 13/29, 17/31)$ and 40-digit arithmetic. The theta tests retain total level at most 3 and both $f = 0, 1$ in both $\eta\eta' = \pm 1$ sectors, with $\eta = 1$. The negative sector uses the diagonal insertion; the ordinary enlarged block in that sector is separately compared with zero.

Fresh comparison	Components	Maximum discrepancy
Physical residue Ward identities	4608	3.25×10^{-41}
Auxiliary residue Ward identities	396	1.73×10^{-41}
Independent physical form vs production	512	3.45×10^{-41}
Independent auxiliary form vs production	44	2.30×10^{-41}
Both embedded Virasoro Ward identities	1536	3.01×10^{-31}
Ramond odd-ground identity	512	5.74×10^{-41}
All-NS physical / auxiliary residues	450 / 180	$3.45 \times 10^{-41} / 0$
Theta: recovered CCY vs physical PBW	960	7.32×10^{-28}
Theta: CCY numerator vs enlarged PBW	960	2.20×10^{-27}
Theta: direct auxiliary sewing vs production	960	1.75×10^{-41}
Theta: enlarged PBW vs convolution	960	4.60×10^{-41}
Theta: odd-loop block without insertion vs zero	480	2.87×10^{-41}

The residue inputs have total level at most 2, with $m, n \in \{-1, 0, 1\}$; positive and negative modes in the identities create the further states required by the check. The Virasoro test covers both copies, L_{-1}, L_{-2} in each slot, $n_1 = 0, 1/2$, $n_2, n_3 = 1/4, 3/4$, both ground parities, and all f, η . The theta block suite took 1.42 seconds, excluding compilation.

The updated graph driver was also run afresh to total level 1 with even NS highest-weight vectors. All 64 Mercedes label choices pass: 53248 recovered components agree with physical PBW to 3.29×10^{-31} , and 4096 unequal-split components satisfy convolution to 8.81×10^{-31} . All eight

Ramond/Ramond glasses choices pass, including zero, one, and two insertions: 320 recovered components agree to 1.32×10^{-30} , and 128 unequal-split components agree to 8.82×10^{-31} . These graph runs took 2.30 and 0.35 seconds. Their cutoff does not activate CCY residues; it tests the uncanceled local signs, full split convolution, and recovery on these graphs. The theta level-3 run tests the CCY recursion as well.

The two production executables also agree in machine precision with each edge cut off at level 1, for both f and all four pairs (η, η') : all 768 components pass, with maximum discrepancy 2.72×10^{-13} . This separate command-line check verifies the normal production entry points. Its eight comparisons took 0.13 seconds including process startup.

Sources, coefficients, results, and reproduction commands are saved in

C++/experiments/coherent_conventions_2026-09-22/.

These are low-level convention checks, not new high-level benchmarks.

9.2 Complete total level six: all 96 channel cases

Every channel in the following table was tested through total level six:

$$r_e \geq 0, \quad \sum_e r_e \leq 6, \quad r_e \in \frac{1}{2}\mathbb{Z} \text{ (NS)}, \quad r_e \in \mathbb{Z} \text{ (R)}. \quad (9.1)$$

Thus an individual edge can reach level 6. The momenta and vertex-label choices are specified in the channel descriptions below. Both NS primary parities are included for NS–R–R theta; the other closed-block tests have even intrinsic primary parities. All formal parity components, including constrained zeros, are retained. These complete-domain calculations all use 40-digit arithmetic. Every one of the 96 cases passes.

Channel	Cases	Multidegrees/case	Recovery error	Time (s)
Theta NS–R–R	16	140	2.09×10^{-25}	188.57
Theta all-NS	2	455	9.29×10^{-29}	1.27
Glasses R/R	8	140	2.87×10^{-25}	27.17
Glasses NS/R	4	252	1.34×10^{-25}	4.73
Glasses NS/NS	2	455	1.27×10^{-29}	0.58
Mercedes	64	3906	4.81×10^{-24}	636.82

Recovery errors compare the extracted physical block with independent physical SCA PBW sewing. The times include the independent references, comparisons, and coefficient output; they are validation-suite wall times, not production block benchmarks. Some channels ran concurrently.

This extension includes the full CCY residues and the nonconstant Schottky vacuum. The graph implementation caches pole geometry, fusion polynomials, global vertices, and repeated recursion states. The universal vacuum is evaluated from primitive Schottky cycles of the ordered graph. NS–R–R vertices reuse the existing recursive branching tables, and inserted vertices reuse the middle branching coefficients. The all-NS vertices additionally require $n = \pm 1, \pm \frac{3}{2}$: their primary vectors are constructed in free fields, converted to physical-PBW/auxiliary coordinates, and evaluated by the local three-point form. These local coefficients are cached. The physical reference uses full inverse Gram matrices; in particular, all three Mercedes spoke metrics act on the central tensor, including their off-diagonal entries.

For glasses and Mercedes, full split powers are retained in the domain

$$\sum_{e:\text{unsplit}} r_e + \sum_{e:\text{split}} \max(r_{e_L}, r_{e_R}) \leq 6. \quad (9.2)$$

Its diagonal is exactly (9.1), and it also contains every expanded-graph multidegree whose ordinary total level is at most 6. The unequal-power comparisons include nonzero modes of $\psi(1)$. Theta retains its established split domain $2r_1 + r_L + r_R + 2r_3 \leq 12$ for $q_1^{r_1} q_L^{r_L} q_R^{r_R} q_3^{r_3}$, with half-integer NS powers and integer Ramond powers.

All 136 NS–R–R theta comparisons pass tolerance 10^{-20} . Each diagonal comparison covers 17920 components. The recovered-block error is tabulated above; the restored double Virasoro numerator agrees with direct enlarged PBW within 3.24×10^{-25} . Direct enlarged recovery agrees with physical PBW within 6.83×10^{-38} , and the graph convolution agrees with enlarged PBW within 9.80×10^{-39} . All spin-lift evaluations agree within 2.50×10^{-25} and the physical loop-sector identity within 1.51×10^{-37} . The 66304 full-split components agree with directly sewn split PBW within 4.02×10^{-22} for double Virasoro and 1.30×10^{-38} for the explicit convolution. The split fermion and enlarged diagonals agree exactly with their zero-mode counterparts (7616 and 8960 components, respectively). The 8960 uninserted obstructed components vanish within 1.37×10^{-37} .

The other graph suites pass tolerance 10^{-18} . Their forward convolution, spin-lift, and independent split checks are:

Channel	Diagonal convolution	All spin lifts	Unequal split powers
Theta all-NS	4.34×10^{-29}	9.29×10^{-29}	—
Glasses R/R	1.83×10^{-24}	5.73×10^{-25}	9.84×10^{-24}
Glasses NS/R	8.75×10^{-26}	1.77×10^{-25}	4.35×10^{-24}
Glasses NS/NS	3.93×10^{-30}	1.27×10^{-29}	—
Mercedes	1.44×10^{-23}	6.79×10^{-24}	2.66×10^{-23}

Each diagonal or spin-lift comparison covers 7280, 8960, 8064, 7280, and 15998976 components, respectively. Unequal split powers add 36736 glasses R/R, 10752 glasses NS/R, and 11845632 Mercedes components. In Mercedes each inserted case has 9690 full-split multidegrees, of which 5784 are off the diagonal. The 32 ordinary and 32 inserted Mercedes cases both have maximum recovery error below 4.81×10^{-24} . Uninserted obstructed numerators vanish within 1.40×10^{-36} , 3.90×10^{-37} , and 1.96×10^{-36} for glasses R/R, glasses NS/R, and Mercedes, respectively.

A separate audit of the saved physical coefficients checks both the full multidegree sets and the physical loop-sector identity, without recomputing any block. It finds no missing or duplicated diagonal multidegree and verifies the independent split-domain counts. The loop identities agree within 1.78×10^{-36} for glasses R/R, 1.47×10^{-36} for glasses NS/R, and 1.06×10^{-38} for Mercedes. To verify the new 40-digit NS Ward implementation against its existing independent Python implementation, all 910 all-NS theta case/multidegree pairs are also compared with `DirectThetaOracle`. This separate machine-precision reference agrees within 5.78×10^{-12} , at tolerance 10^{-9} ; it does not determine the errors in the 40-digit tables above.

Before the complete runs, the extended implementation also passed total-level-two checks for all-NS theta (two cases, 35 multidegrees each), glasses R/R (four cases, 14 multidegrees each), and Mercedes (four cases, 71 multidegrees each). Their saved results are retained with the extension.

The extension exposed a sign error in the new graph CCY driver that the initial cutoff could not detect. For a self-loop sewn between local slots (1, 0), the residue at null level rs needs the factor $(-1)^{rs}$ from the middle-slot Virasoro Ward normalization. Omitting it first produces a discrepancy at physical multidegree (0, 0, 3) in glasses. All glasses results above are from fresh runs after correcting that factor. The Mercedes graph has no self-loops, so its concurrent run does not execute the corrected branch. No production header or manuscript formula was changed by this test-driver correction.

The Mercedes run sums 29240 branching tuples and makes 896784 CCY recursion transitions. Its double Virasoro stage took 371.45 seconds, direct auxiliary sewing 0.20 seconds, and physical PBW sewing 29.80 seconds across all cases. The remaining time is coefficient comparisons, recovery, spin

evaluations, and output. It saves all 249984 diagonal coefficient rows, with physical, enlarged, and recovered values.

All extension sources, results, coefficients, and reproduction commands are in `C++/experiments/total_level6_2026-09-16/`. The `README.md` gives the cutoffs and implementation details; `manifest.json` summarizes the results and source hashes. The script `run_all.py` performs a fresh sequential reproduction. The saved-data audit is `coverage_and_sector_results.json`; the separate NS Ward-port check is `ns_physical_port_results.json`. The diagnostic self-loop failures are retained separately and are not counted as passing results. These tests use the NS/R assignments and primary-parity cases listed below, at the same generic momenta; they do not add the inequivalent four-edge Ramond cycle on the Mercedes graph or singular limits.

9.3 Initial NS–R–R theta tests, including independent split powers

The NS–R–R theta tests use 40-digit arithmetic. We take $b = 7/5$ and graph momenta $(11/23, 13/29, 17/31)$. Both $p_1, f = 0, 1$, all four vertex-sign pairs, every formal parity component, and all eight evaluations of the spin lifts are covered, giving 16 cases. The diagonal physical cutoff is total level 3: 30 multidegrees and 240 parity components per case. The independent split series retains $q_1^{r_1} q_L^{r_L} q_R^{r_R} q_3^{r_3}$ with $2r_1 + r_L + r_R + 2r_3 \leq 6$, half-integer NS powers, and integer Ramond powers; its diagonal is the same physical cutoff. There are 130 multidegrees and 1040 components per inserted case. The full split numerator is checked against an independently sewn enlarged PBW numerator, including the modes of $\psi(1)$ that change the middle level. The double Virasoro calculation uses both CCY recursions and the Schottky vacuum. The numerical insertion is $Q\Theta\psi$ in both the numerator and auxiliary factor.

Comparison	Components	Maximum discrepancy
Recovered double Virasoro vs physical PBW	3840	7.32×10^{-28}
Double Virasoro numerator vs enlarged PBW	3840	2.20×10^{-27}
Recovery from enlarged PBW vs physical PBW	3840	1.01×10^{-40}
Graph convolution vs enlarged PBW	3840	7.73×10^{-41}
Recovered block vs PBW at all spin lifts	3840	1.35×10^{-27}
Physical loop-sector identity	3840	1.04×10^{-40}
Split fermion diagonal vs zero-mode factor	1536	0
Split enlarged diagonal vs zero-mode enlarged PBW	1920	0
Full split double Virasoro vs split enlarged PBW	8320	7.06×10^{-26}
Full split convolution vs split enlarged PBW	8320	6.29×10^{-41}
Uninserted enlarged PBW in obstructed cases vs zero	1920	4.34×10^{-40}

All 136 comparisons pass tolerance 10^{-20} . The two diagonal/zero-mode identities agree exactly in the saved arithmetic. The 16 fresh numerical processes took 3.61 seconds together; this is a validation-suite time, not a high-level block benchmark.

9.4 Initial all-NS theta and local all-NS vertices

Exact rational tests cover all eight intrinsic-primary-parity assignments and all eight branch-label tuples $n_i \in \{0, \frac{1}{2}\}$. All 64 local vertex comparisons and all 384 L_{-1} Ward identities (three slots in each Virasoro copy) pass. These archived runs also included intrinsic odd NS primaries, whereas the conventions specified here take NS highest-weight vectors even.

The independent closed all-NS theta calculation has even primaries and total level 2. All 35 coefficients agree with physical PBW within 4.67×10^{-15} . All eight spin-lift choices in both vertex-parity sectors give 16 evaluation checks, with maximum discrepancy 5.56×10^{-17} . The tolerance is

10^{-10} . The series backend uses machine arithmetic, with 40-digit branching-polynomial evaluation; this is not a 40-digit end-to-end calculation.

9.5 Initial glasses tests with zero, one, or two insertions

The glasses tests use the same momenta (bridge, left handle, right handle), machine arithmetic, even intrinsic primary parities, and individual edge level 1. They cover R/R, NS/R, and NS/NS handles; R/NS is the isomorphic assignment obtained by exchanging the named handles, rather than a separate numerical run. Both vertex-form parities, all Ramond signs, and every spin-parity slot are included: eight R/R cases, four NS/R cases, and two NS/NS cases. The R/R tests include zero, one, and two inserted loops; 960 of the full-split components have unequal powers on a split pair. At this cutoff every Virasoro edge has descendant level at most one, so the exact CCY answer equals its global term. These glasses tests do not test higher-level residues or nonconstant Schottky corrections.

Comparison	Components	Maximum discrepancy
R/R diagonal numerator vs PBW convolution	768	7.02×10^{-13}
R/R recovery vs physical PBW	768	8.71×10^{-14}
R/R full split numerator vs PBW convolution	1728	8.30×10^{-13}
NS/R numerator vs PBW convolution	576	3.17×10^{-14}
NS/R recovery vs physical PBW	576	2.79×10^{-14}
NS/NS numerator vs PBW convolution	432	1.45×10^{-15}
NS/NS recovery vs physical PBW	432	1.90×10^{-15}

The uninserted enlarged block is separately checked to vanish in every obstructed R/R case, with maximum absolute residual 1.17×10^{-13} . All checks pass tolerance 10^{-9} . The R/R diagonal, full-split, and NS-assignment tests record runtimes of 0.079, 0.105, and 0.042 seconds, respectively, excluding compilation.

9.6 Initial Mercedes tests: ordinary and obstructed sectors

For the graph in (7.5), take $b = 7/5$ and edge momenta $(11/23, 13/29, 17/31, 19/37, 23/41, 29/43)$. Every original edge has an independent physical cutoff of level 1. The three NS powers are $0, \frac{1}{2}, 1$, and the three Ramond powers are $0, 1$, giving $3^3 2^3 = 216$ multidegrees per case, including total level 6. The inserted test also retains unequal split powers $(0, 1)$ and $(1, 0)$, with each segment independently bounded by level 1. Thus each inserted case has 432 full-split multidegrees, half of them diagonal.

All eight choices of (f_1, f_2, f_3) and all eight choices of $(\eta_{v_1}, \eta_{v_2}, \eta_{v_3})$ are tested. Equation (7.6) gives 32 ordinary and 32 obstructed cases. Every case retains all 64 formal parity components and compares all 64 evaluations of the edge spin lifts. Intrinsic primary parities are even. Arithmetic is complex machine precision; the central NS branching polynomials are evaluated at 40 digits before conversion.

The physical reference contracts SCA PBW vertices and inverse SCA Gram matrices. Its Ramond vertices use `ScaWard`, and its central NS vertex uses the independent `NSDescendantThreeForm`; it does not use branching data. The enlarged numerator instead uses `LowAnchors` at the three NS-R-R vertices, `ns_fusion_data` at the center, and `MiddleBranching` on the split edge, followed by two Virasoro graph contractions. Local vertex and branching values are reused within the run. At this cutoff every Virasoro descendant edge has level at most one, so its exact CCY answer is the global term, evaluated with the L_{-1} Ward formulas. No higher Kac residue or nonconstant Schottky correction enters this test.

The auxiliary factor is sewn directly from fermion Fock states and Ward forms. Recovery uses only the double Virasoro numerator and this factor: it subtracts lower-level convolutions, divides by

the matching-sector constant. Physical PBW coefficients are used only for comparison. Unequal split powers are compared with the independent physical-PBW/fermion convolution, including nonzero fermion modes. Both inserted factors use $Q\Theta\psi$, so the normalization cancels upon recovery.

Comparison	Ordinary	Inserted
Recovered physical block vs PBW	1.55×10^{-12}	1.60×10^{-12}
Diagonal numerator vs PBW convolution	1.26×10^{-12}	2.31×10^{-12}
Physical loop-sector identity	6.67×10^{-16}	6.67×10^{-16}
Recovered block vs PBW at all spin lifts	2.14×10^{-12}	2.20×10^{-12}
Unequal split powers vs PBW convolution	—	2.03×10^{-12}

Each of the first four rows covers 442368 components per column: $32 \times 216 \times 64$. The recovery comparisons contain 6912 nonzero formal components per column; constrained zeros are retained. The unequal-split check adds 442368 components. The uninserted enlarged blocks in all 32 obstructed cases vanish with maximum absolute residual 9.14×10^{-13} , while their physical blocks are generally nonzero. All checks pass tolerance 10^{-8} .

The new central-fermion adapter initially omitted the bra-slot BPZ sign. The raw sphere Pfaffian needs the factor $(-1)^{\epsilon'_1}$, consistently with the auxiliary $\psi_{-1/2}$ norm -1 used above. Correcting that adapter resolved the mismatch; the graph formulas and production headers were unchanged. The results tabulated here use the corrected convention.

The C++ test process took 1.93 seconds of wall time, including coefficient output; its internal timer recorded 1.59 seconds. Central-data export took 0.098 seconds and compilation 3.29 seconds, for a fresh total of 5.32 seconds. These are low-level validation timings. The run saves all 13824 diagonal coefficient rows, each with the physical PBW, enlarged double Virasoro, and recovered parity arrays.

9.7 Local Ramond identity and external-state projections

The archived Ramond odd-ground check covered 864 local Ward tests at 40 digits. The fresh check of (5.1) in the current normalization is recorded in the first subsection. These cover $p, f = 0, 1$, both vertex signs, the NS primary and its $G_{-1/2}$ and L_{-1} descendants, and both Ramond grounds and their G_{-1} and L_{-1} descendants in each Ramond slot. The maximum discrepancy is 2.90×10^{-41} .

At $b = 7/5$, $P = 13/29$, 40-digit external-state BPZ projections reconstruct every tensor-PBW input at NS levels $0, \frac{1}{2}, 1$ and R levels $0, 1$:

Sector, level	Inputs	Gram entries	Gram error	Reconstruction error
NS, 0	1	1	0	0
NS, $\frac{1}{2}$	2	4	1.40×10^{-32}	5.68×10^{-33}
NS, 1	2	4	1.42×10^{-32}	3.62×10^{-33}
R, 0	4	16	5.86×10^{-33}	2.93×10^{-33}
R, 1	12	144	1.27×10^{-30}	1.29×10^{-30}

This gives 21 independent inputs and 169 branch-Gram entries, including all four Ramond ground tensors. Branch vectors are obtained from free-field primary and generator actions. Their coordinate conversion to the physical-PBW/auxiliary basis has maximum residual 2.70×10^{-32} and is used only to evaluate BPZ overlaps. The final reconstruction uses independently known inverse Virasoro Gram matrices, not a fit of the input to the branch basis. This checks the external-state expansion locally, not a full correlator with external insertions.

9.8 Exact graph-sign and parity-algebra checks

Exact operator-order permutations verify (8.3), while independent parity enumeration checks the associativity of the graph convolution:

Graph	Parity pairs	Associativity triples
Theta	64	512
Glasses	64	512
Local ordering with external states	4096	262144

The theta driver also checks all eight half-edge parity assignments and all 64 explicit convolution-sign pairs. Four loop cases check commutation of the cycle monomials and the squares of the normalized cycle terms in (5.6). All finite checks pass exactly. The external-state row is a local ordering check before sewing the NS slots; it is not a separate recovery construction. The Mercedes calculation retains its central sphere and uses the single external insertion described above.

9.9 All-NS Mercedes: comparison with NS c -recursion

We also compare the all-NS Mercedes block with the NS c -recursion in the SCblock notes, independently of physical PBW sewing. All six edges are NS, with even intrinsic primary parities. Use the vertex orders in (7.5), $b = 7/5$, and momenta (11/23, 13/29, 17/31, 19/37, 23/41, 29/43). Both calculations use 40-digit arithmetic and retain every nonnegative half-integer multidegree with total level at most 6: 18564 multidegrees. All eight allowed vertex-form assignments are included. At fixed NS multidegree the edge parities are fixed, so coefficientwise agreement also covers all 64 spin-lift choices.

The double-Virasoro side uses both full CCY recursions and Schottky vacuum factors, then removes the directly sewn auxiliary fermion factor. The independent NS recursion uses the Kac poles, inverse null slopes, fusion polynomials, and explicit global $\mathfrak{osp}(1|2)$ vertices. Its vacuum seed is expanded from the primitive Schottky cycles of this graph. Through level 6, the four triangles contribute supercurrent terms at levels 9/2, 11/2 and stress-tensor terms at level 6; the three four-edge cycles first contribute at level 6. Thus this comparison includes the nonconstant vacuum seed.

The vacuum/global product requires the full sign in (3.2), with ϵ the global parity and ϵ' the vacuum parity. In particular, moving the middle-slot odd global operators past the ket-slot odd vacuum operators contributes $(-1)^{\sum_v \epsilon_{v,2} \epsilon'_{v,3}}$ in addition to the graph permutation sign. This term cancels in theta but survives in Mercedes. Omitting it gives 130 discrepant coefficients, beginning at total level 5; including it gives agreement throughout the requested domain. The maximum scaled error is 1.05×10^{-28} and the maximum absolute error is 1.18×10^{-28} .

The fresh run takes 15.78 seconds: 10.03 seconds for the enlarged double-Virasoro blocks, 0.013 seconds for the auxiliary factor, 0.162 seconds for recovery, and 5.58 seconds for NS recursion, comparisons, and output. Compilation is excluded. Sources, coefficients, the diagnostic with the omitted sign, and reproduction instructions are saved in `C++/experiments/mercedes_all_ns_c_recursion_2026-09-16/`. This additional test covers the all-NS assignment with even intrinsic primaries; it does not extend the tested mixed NS/R assignments.

Extension to total level 10. At the same parameters and precision, the two methods also agree on all 230230 multidegrees through total level 10. The all-even vertex-form sector contains 29029 multidegrees, and each of the other seven sectors contains 28743. The maximum scaled error is 2.05×10^{-25} and the maximum absolute error is 4.00×10^{-25} ; no coefficient fails the 10^{-18} tolerance. No additional physical PBW block is computed.

This extension uses the full truncated Schottky product: 13 primitive unoriented cycles of lengths at most six and 20 fermionic oscillator factors, including their products. The lift phases are transported under cycle composition, and each half-multiplier is checked to square to the corresponding multiplier in the cycle convolution algebra. The chosen coordinate matrices give edge-map determinant $+q_e$ when exactly one endpoint is an infinity slot, and $-q_e$ otherwise. This replaces the old generic helper’s uniform $-q_e$ assumption; its first effect on the bosonic vacuum is at total level 9, so the level-six result is unaffected.

The fresh calculation retains 45405 branch tuples and performs 10921308 CCY residue transitions. It generates the residual Virasoro level domain directly after subtracting each branch’s starting levels, and convolves only pairs within that domain. Branching data and local recursion data are reused. Recovery updates only the higher coefficients that receive a nonconstant auxiliary contribution. On the NS recursion side, the full-block cache is cleared between target multidegrees: accumulated shifts plus remaining levels equal the target, so those entries cannot be shared by different targets. The reusable local caches are retained.

Stage, 40-digit arithmetic	Time (s)
Double-Virasoro enlarged blocks	681.135
Direct auxiliary factor	0.183
Collect enlarged coefficients	0.071
Recovery and coefficient output	1.892
Full double-Virasoro pipeline	683.282
NS Schottky seed	0.178
NS recursion and coefficient output	270.148
Full NS recursion stage	270.326

The two sequential computational stages take 953.61 seconds in total, excluding compilation and the subsequent comparison of saved coefficients. Both coefficient sets, errors at each total half-level, source hashes, and reproduction instructions are saved in `C++/experiments/mercedes_all_ns_level10_2026-09-16/`.

9.10 Saved evidence, reproduction, and scope

The theta, glasses, local Ward, external-state, and finite graph scripts, commands, source hashes, and results are saved in `C++/experiments/arbitrary_plumbing_2026-09-16/`. The reproduction guides are `theta_README.md` and `glasses_README.md`; the directory’s `README.md` maps the formulas to their checks. The result files are `theta_summary.json`, `theta_all_ns_results.json`, `theta_ramond_local_results.json`, `external_state_results.json`, `glasses_results.json`, `glasses_full_split_results.json`, `glasses_ns_assignments_results.json`, and `graph_sign_results.json`.

The Mercedes sources and results are saved separately in `C++/experiments/genus3_mercedes_2026-09-16/`. Running `python3 run.py` from that directory exports the central data, compiles the C++ driver, and executes the complete 64-case suite. The `manifest.json` records parameters, graph ordering, source and result hashes, timings, and aggregate errors. The `results.json` records every case and its worst discrepancies; `coefficients.jsonl` contains the diagonal coefficient arrays, and `central_manifest.json` records the independent central-vertex data. Its `README.md` explains the conventions and reproduction commands.

These checks concern generic momenta at the stated cutoffs. For Mercedes they cover the all-NS and triangular Ramond assignments, excluding the inequivalent four-edge Ramond cycle and intrinsic-odd NS primaries. They do not test singular weights, full external-Ramond correlators, or

a SCFT spectral sum or moduli integral. No arbitrary-genus numerical frontend is claimed: the general argument uses the local identities and graph contractions above.